

$$\bar{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

both complex number

$$S = I_{\text{rms}}^2 Z = \frac{V_{\text{rms}}^2}{Z^*} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

+ve

1. $Q = 0$ for resistive loads (unity pf).
2. $Q < 0$ for capacitive loads (leading pf).
3. $Q > 0$ for inductive loads (lagging pf).

$$\text{Complex Power} = S = P + jQ = \mathbf{V}_{\text{rms}} (\mathbf{I}_{\text{rms}})^*$$

$$= |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| \angle \theta_v - \theta_i$$

$$\text{Apparent Power} = S = |\mathbf{S}| = |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2}$$

$$\text{Real Power} = P = \text{Re}(\mathbf{S}) = S \cos(\theta_v - \theta_i)$$

$$\text{Reactive Power} = Q = \text{Im}(\mathbf{S}) = S \sin(\theta_v - \theta_i)$$

$$\text{Power Factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

The **apparent power** (in VA) is the product of the rms values of voltage and current.

Complex power (in VA) is the product of the rms voltage phasor and the complex conjugate of the rms current phasor. As a complex quantity, its real part is real power P and its imaginary part is reactive power Q .

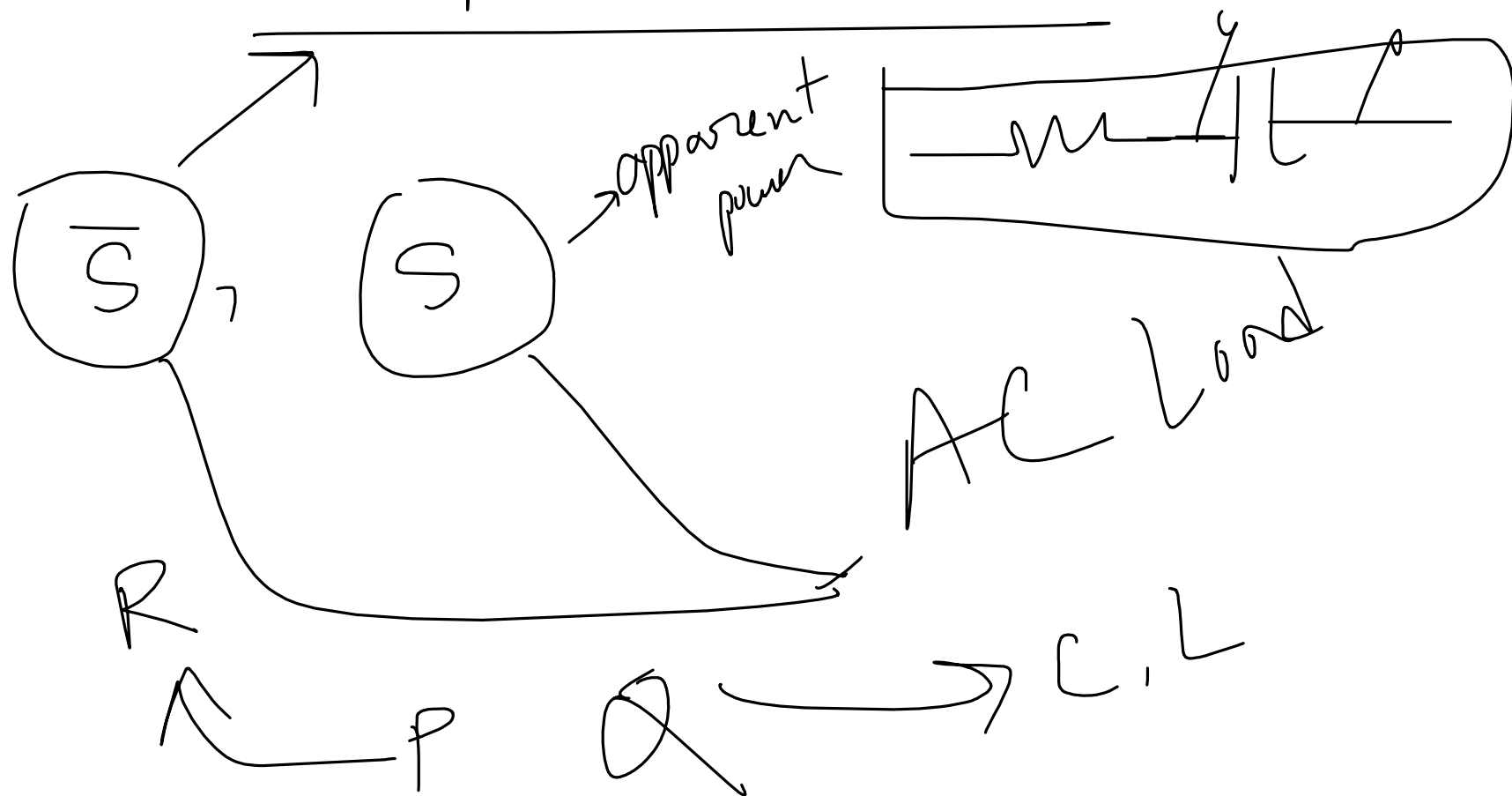
$$S = V_{\text{rms}} I_{\text{rms}}$$

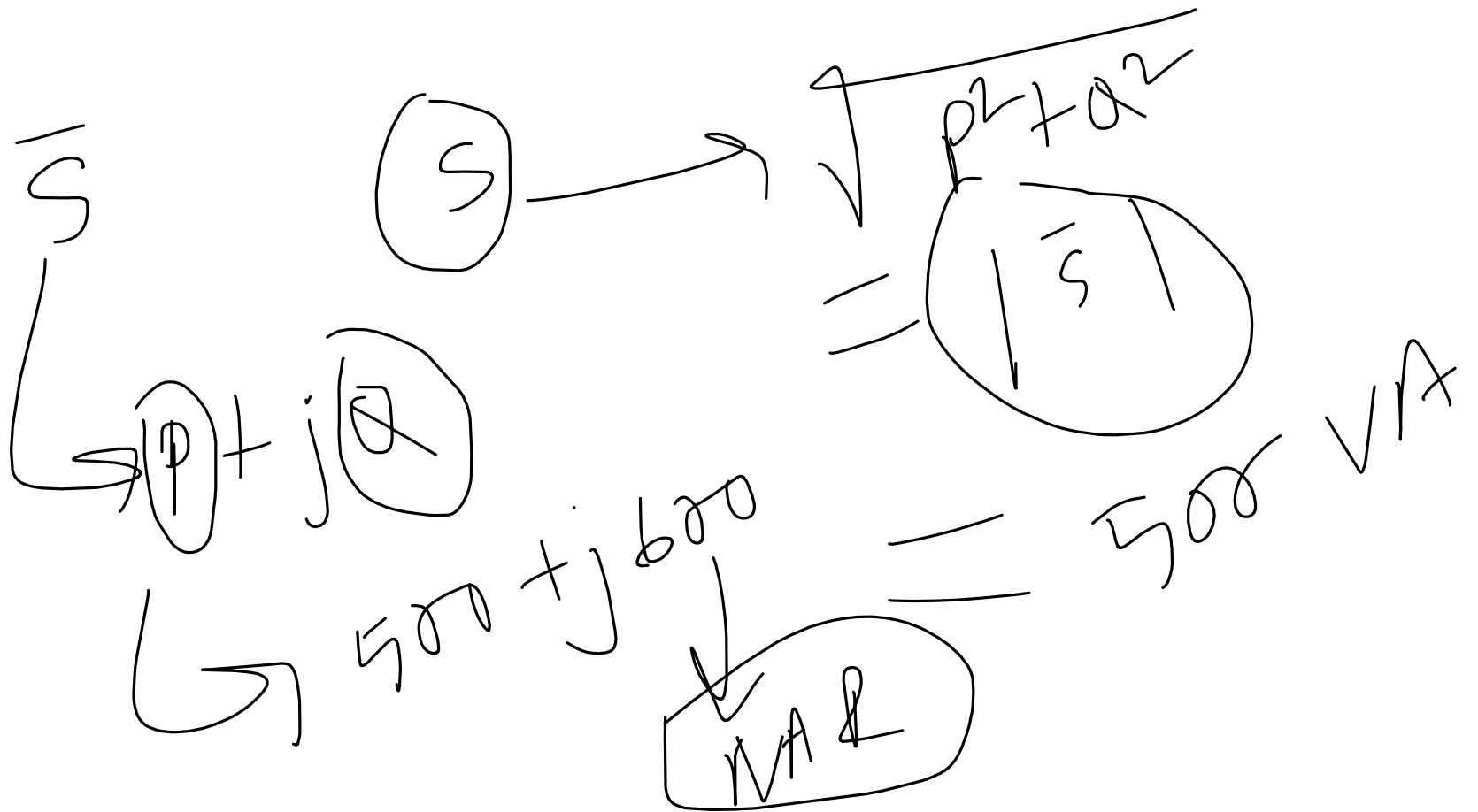
scalar

apparent

$$\bar{S} = P + jQ$$

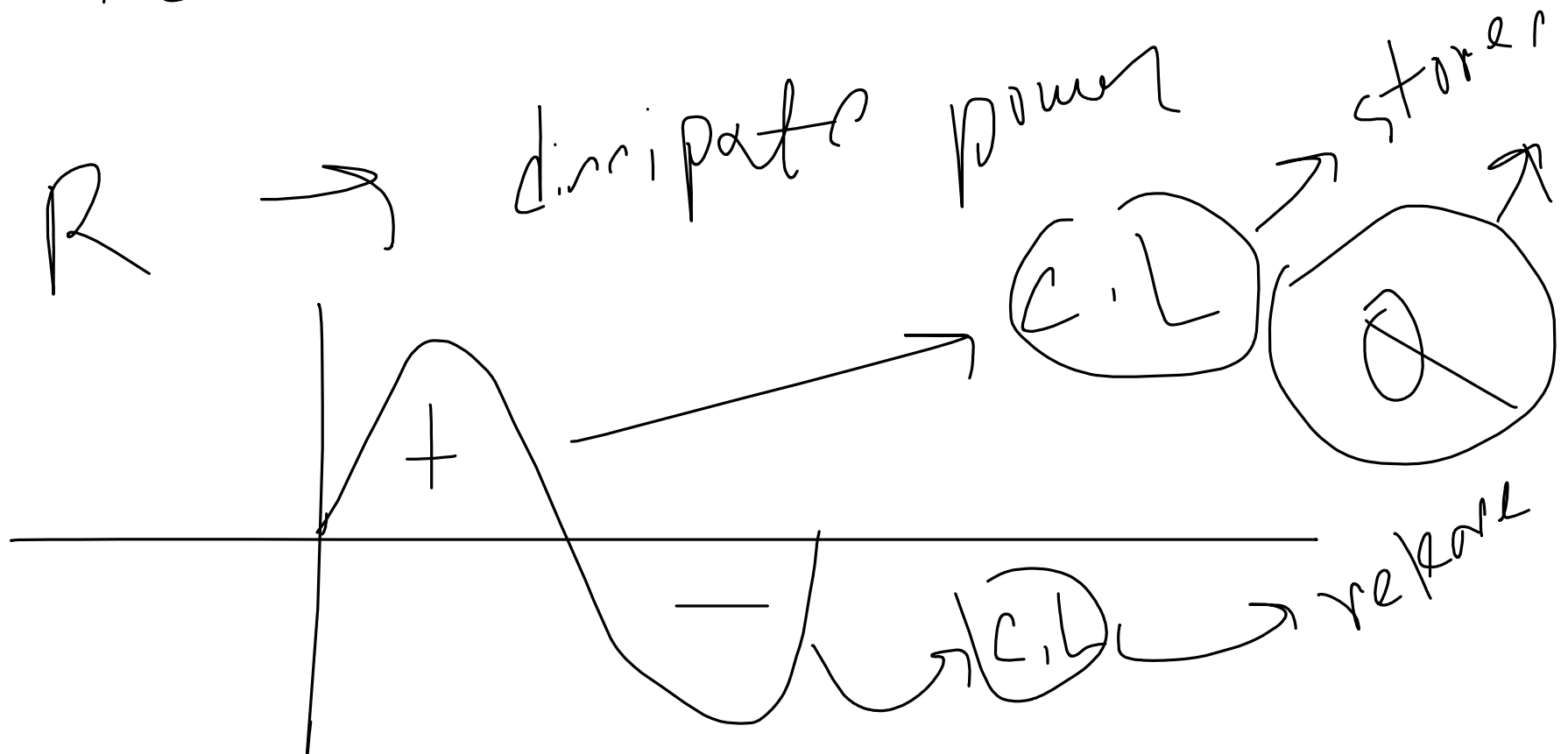
Complex Power



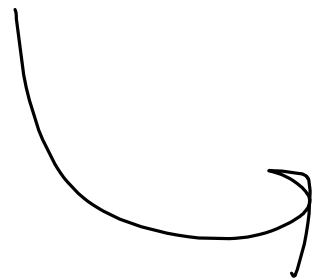


$L, C \rightarrow$ not even use power

$R \rightarrow$ dissipate power



$$PF = \left[\frac{P}{S} \right]$$



how much

useful $\times^n$
energy in the sys

$$0.9$$

$\phi =$ work

pf correction



For a load, $\mathbf{V}_{\text{rms}} = 110 \angle 85^\circ \text{ V}$, $\mathbf{I}_{\text{rms}} = 0.4 \angle 15^\circ \text{ A}$. Determine: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

(a)

$$\begin{aligned}\bar{S} &= \bar{V}_{\text{rms}} \times \bar{I}_{\text{rms}} \\ &= (110 \angle 85^\circ) \times (0.4 \angle -15^\circ) \\ &\rightarrow 15 + j41.35 \\ S &= |\bar{S}|\end{aligned}$$

(b)

$$S = 15 + j41.35$$
$$= P + jQ$$

- 1. $Q = 0$ for resistive loads (unity pf).
- 2. $Q < 0$ for capacitive loads (leading pf).
- 3. $Q > 0$ for inductive loads (lagging pf).

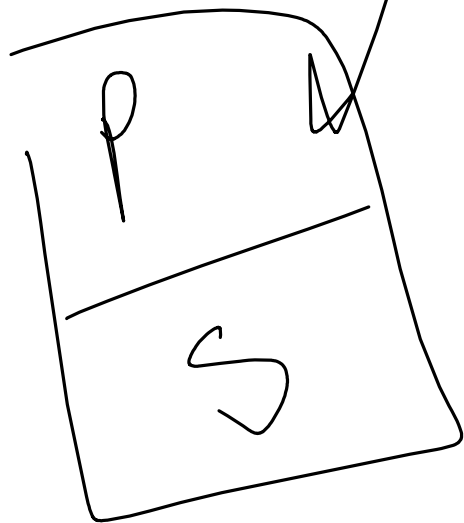
$$Q = 41.35 (+ve)$$

↳ inductive load
lagging pf.

(C)

~~(B)~~

$$pf = \cos(\theta_v - \theta_i)$$
$$= \cos(85^\circ - 1^\circ)$$



$$= 0.34$$

A hand-drawn diagram of a rectangular box containing the text "0.9" followed by an upward-pointing arrow. An arrow points from the box containing "0.34" to this box.

$$Z = \frac{V_{rms}}{I_{rms}}$$

A sinusoidal source supplies 100 kVAR reactive power to load $Z = 250 \angle -75^\circ \Omega$. Determine: (a) the power factor, (b) the apparent power delivered to the load, and (c) the rms voltage.

Practice Problem 17.12

$$Q = 100 \text{ k}$$

$$S = P + jQ$$

$$P = S \cos \theta$$
$$Q = S \sin \theta$$

The **power factor** is the cosine of the phase difference between voltage and current. It is also the cosine of the angle of the load impedance.



$$Z = 25 \angle -75^\circ$$

$$pf = \frac{P}{S} = \cos(\theta_v - \theta_i)$$
$$= \cos(\theta_z)$$

$$\therefore pf = \cos(-75^\circ) = \boxed{0.26}$$

$$Q = S \sin \theta$$

$$S = \frac{Q}{\sin \theta}$$

$$= \frac{100 \text{ V}}{(-0.97)}$$

$$S = -103 \text{ V}$$

$$\sin(-75^\circ)$$

$$\underline{S} = P + jQ$$

$$= S_{\text{env}} + j \times 100k$$

$$= 26678 + j100k$$

$$S = I_{\text{rms}}^2 Z = \frac{V_{\text{rms}}^2}{Z^*} = V_{\text{rms}} I_{\text{rms}}^*$$

$$V_{\text{rms}} S =$$

$$S = \boxed{V_{\text{rms}} \times I_{\text{rms}}}$$

$$S = \frac{V_{\text{rms}}^2}{Z^*}$$

$$\S \quad \tilde{V}_{rms} = \frac{\tilde{S}}{Z^*}$$

$$= \frac{26.66k + j100k}{250 \angle -1.75^\circ}$$

$$pf_1 = 0.34$$

$$pf_2 = 0.26$$

(1)

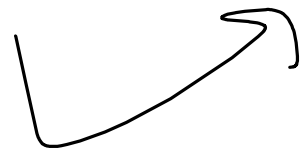
$$pf = \frac{p}{s}$$

~~0~~ → X

pf → 0.9 ↓ ↪ wantful energy ↑

changing from low pf to high pf system voltage & current without altering correction

high pf

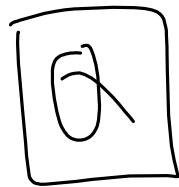


the system have

pf high means more



than



~~Q~~ $\rightarrow$ we don't want. wasteful energy.

$$P = S_1 \cos \theta \quad \text{--- (1)}$$

$$Q = S_1 \sin \theta \quad \text{--- (2)}$$

$$(2) - (1) \Rightarrow$$

$$\frac{S_1 \sin \theta}{\cos \theta} = \frac{Q}{P}$$

$$\therefore Q = P \tan \theta$$

$$Q_2 = P \tan \theta_2$$

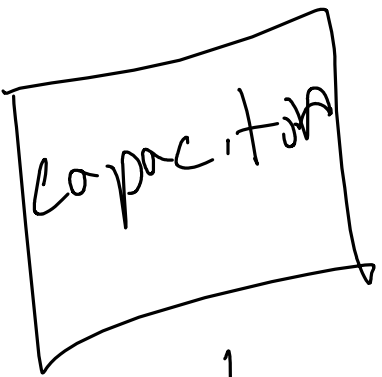
$$\begin{aligned} Q_c &= Q_1 - Q_2 \\ &= P (\tan \theta_1 - \tan \theta_2) \end{aligned}$$

$$\begin{aligned} P &= V \times I \\ &= \frac{V^2}{R} \end{aligned}$$

$$\begin{aligned} Q_c &= \frac{V_{rms}^2}{X_c} \\ Q_c &= \frac{V_{rms}^2}{1/\omega C} \end{aligned}$$

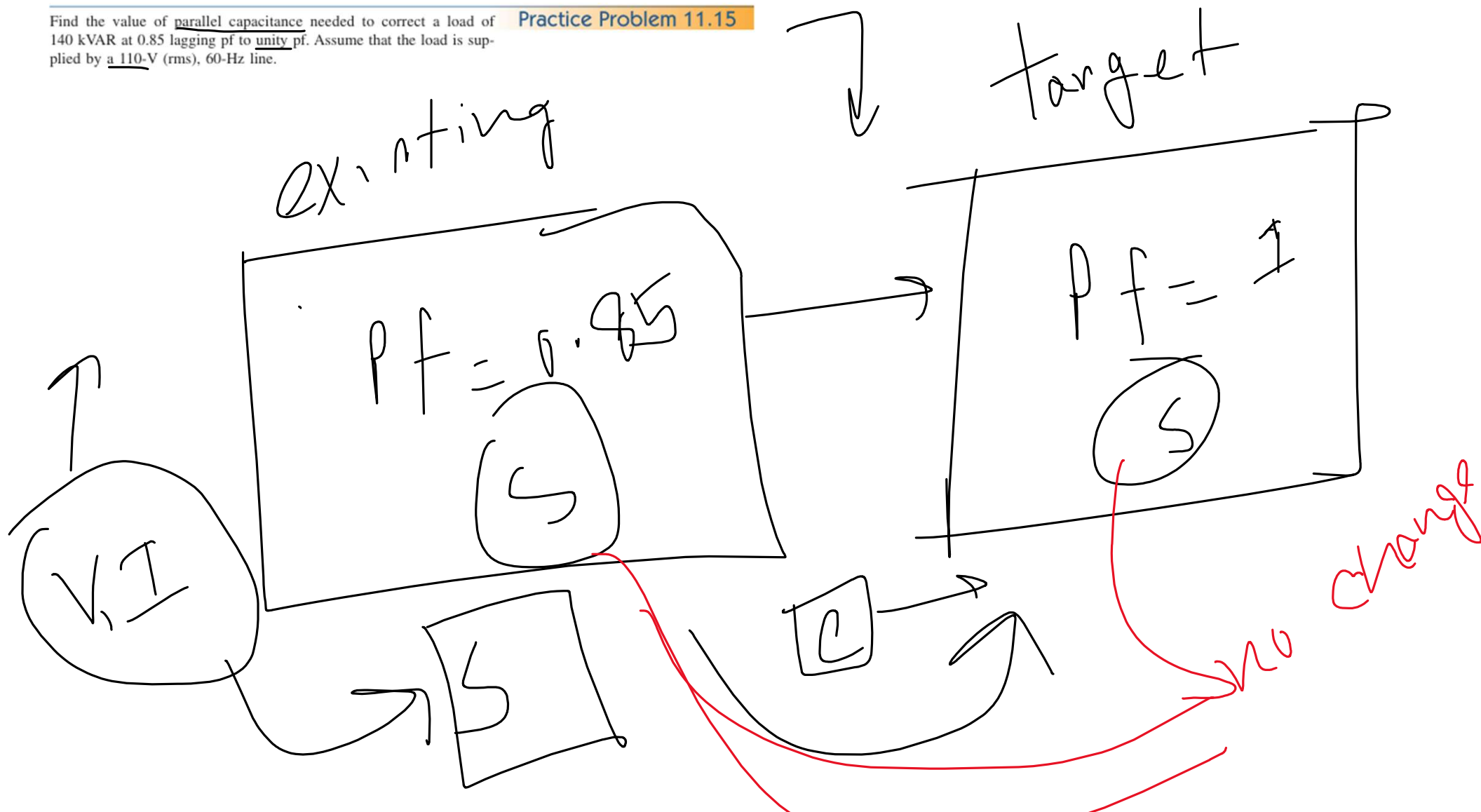
$$Q = \frac{V}{X_c}$$

Mechanism

we connect  in parallel
to the load  to change pf.

Find the value of parallel capacitance needed to correct a load of 140 kVAR at 0.85 lagging pf to unity pf. Assume that the load is supplied by a 110-V (rms), 60-Hz line.

Practice Problem 11.15



$$\phi_1 = 140^\circ$$

$$\cos \theta_1 = 0.85$$

$$\theta_1 = 31.79^\circ$$

$$\textcircled{1} \quad \phi_c = \phi_1 - \theta_2$$

$$\textcircled{2}$$

$$\phi_c =$$

$$\theta_2 = \frac{V_{rms}^2}{1/\omega C}$$

$$S = \frac{\cancel{140} \text{ K}}{\sin \theta_1}$$

$$= \frac{140 \text{ K}}{\sin(31.79^\circ)}$$

$$= 265751.75 \text{ VA}$$

Second system

$$\cos(\theta_2) = 1$$

$$\therefore \theta_2 = 0^\circ$$

$$\begin{aligned} Q_2 &= \int \sin \theta_2 \\ &= \int \times 0^\circ = 0 \end{aligned}$$

$$\boxed{Q_L} \geq Q_1 - Q_2$$

$$= 140k - 0$$

$$= \boxed{140k}$$

$$Q_L = \frac{V_{rms}^2}{\frac{1}{\omega C}}$$

$$\omega = 2\pi \times 60$$

$$=$$

$$Q_L = V_{rms}^2 \times \omega C$$

$$C = \frac{Q_L}{V_{rms}^2 \times \omega}$$

$$= \frac{140k}{(110)^2 \times 2\pi \times 60Hz}$$

$$C = 30.69 \text{ mF.}$$

